

Auditable Product-Stable Window Selection for Strong-Motion Records

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Abstract

Strong-motion products depend on the processing interval used to compute peak motion, waveform energy, and duration-sensitive measures. Fixed windows are simple to deploy in batch workflows. Their product retention can change sharply across archives. We evaluate this behavior with 53,463 three-component records from InstanceGM and K-NET. Each candidate window is audited by three checks: retention of full-record peak ground acceleration, retention of relative waveform energy, and inclusion of the full-record peak time. Fixed 42.00 s windows show strong dataset dependence. On InstanceGM, feature-onset, energy-onset, and catalog-P fixed windows have instability rates of 81.25%, 69.91%, and 73.03%. On K-NET, the same windows have instability rates of 4.15%, 3.10%, and 8.25%. We then evaluate a waveform-derived selector that chooses the shortest candidate passing the product checks and assigns the full record only when every shorter candidate fails. Only 0.84% of records are assigned to full-record processing, with median selected-window durations of 84.94 s for

InstanceGM and 24.66 s for K-NET. The selector recovers 25,468 InstanceGM feature-onset fixed-window failures and 21,913 InstanceGM energy-onset fixed-window failures under the same audit. A 5% damping response-spectrum audit gives overall PSA-retention failure rates of 12.98%, 22.26%, and 32.28% for feature-onset fixed windows at 0.2 s, 1.0 s, and 3.0 s. The shortest-stable selector reduces the same rates to 0.02%, 0.87%, and 5.56% (9, 465, and 2,972 records). Threshold tests make the operating trade-off explicit: the 0.95 energy-retention criterion keeps full-record assignment rare, whereas a 0.98 criterion raises the overall assignment rate to 46.84%. The results define a reproducible offline windowing audit for strong-motion product preparation.

Plain Language Summary

Strong-motion records need processing windows that keep the shaking used by engineering and seismological products. A fixed 42.00 s window can lose useful motion in one archive and include more time than needed in another. We test windows by checking whether they keep the full-record peak motion, enough waveform energy, and the time of the full-record peak. A simple offline selector chooses the shortest candidate window that passes these checks. The selector gives longer typical windows for InstanceGM and shorter typical windows for K-NET, with rare full-record assignment at the stated thresholds.

Introduction

Strong-motion records support earthquake engineering, shaking assessment, ground-motion model development, and post-event review. Products such as peak ground acceleration, energy measures, response spectra, and duration-related metrics depend on the part of the record used during processing. Archive-scale pipelines often use fixed-duration windows because they are easy to implement and easy to document. A single fixed interval can miss late product-relevant motion in one archive and retain unnecessary waveform in another.

These dependencies are a central issue in strong-motion processing and record-quality work (Douglas, 2003; Boore and Bommer, 2005; Boore et al., 2012).

Fixed windows are also common in public waveform datasets because they make records uniform, easy to index, and reusable across studies. That practical value creates a product-quality requirement: the retained interval needs to be checked against the strong-motion quantities computed from the record. The record-level question is whether the selected interval preserves peak motion, waveform energy, and spectral amplitudes used by downstream users. This study audits that question directly.

In a production strong-motion workflow, the window is a record-level processing decision made before product tables are exported. The same interval supports peak values, response spectra, energy-like measures, duration measures, and later review of flagged records. A useful selector keeps product-relevant motion, exposes the records that need full-record processing, and leaves an audit trail that can be checked after archive updates. Response spectra and duration measures are standard engineering products, which makes window retention a product-level requirement beyond plotting convenience (Trifunac and Brady, 1975; Dobry et al., 1978; Chopra, 2017).

This study treats processing-window selection as a product-retention problem. Arrival estimates help locate the beginning of shaking. The processing window also retains the quantities used downstream. The target output is an offline record-level window with traceable product checks. This framing matches strong-motion archive preparation and batch product generation, where the full record is available before a final processing window is stored.

We use InstanceGM and K-NET to test the same audit on archives with different product-window behavior. InstanceGM provides a large Italian strong-motion and seismic waveform archive through the INSTANCE data family (Michellini et al., 2021; see Data and Resources). K-NET provides Japanese nationwide strong-motion recordings operated by NIED (Aoi et al., 2004; Okada et al., 2004; see Data and Resources). Both datasets are evaluated at 100 Hz median sampling rate, keeping the comparison focused on waveform scale and window

76 policy. Figure 1 summarizes the workflow.

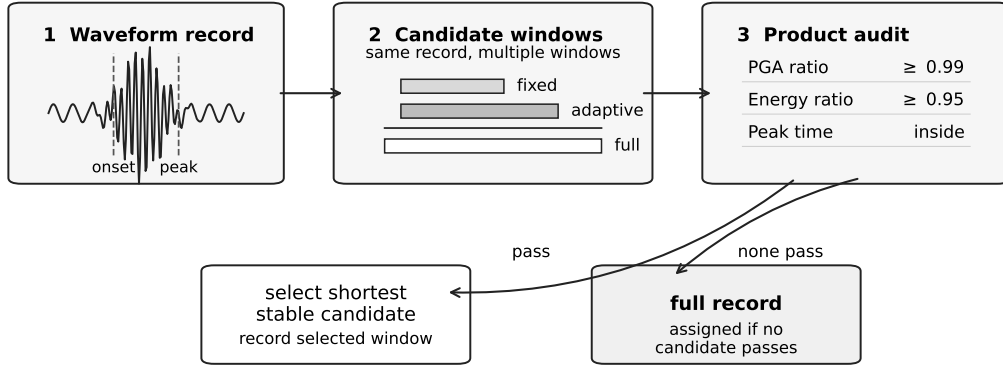


Figure 1: Workflow for offline product-stable window selection. Each waveform record is converted into candidate processing windows. Candidate windows are evaluated by retention of full-record PGA, relative energy, and inclusion of the full-record peak. The selector chooses the shortest stable non-full candidate and assigns the full record when no non-full candidate passes the product-retention checks.

78 The paper makes four contributions. First, it defines product stability for strong-motion
 79 processing windows using peak retention, energy retention, and peak-time inclusion. Sec-
 80 ond, it quantifies the dataset dependence of common fixed windows across 53,463 waveform
 81 records. Third, it evaluates a no-catalog, waveform-derived shortest-stable selector with
 82 explicit full-record accounting. Fourth, it reports product-impact recovery, threshold sensi-
 83 tivity, and response-spectrum retention as auditable engineering-product checks.

84 Data

85 The waveform audit contains 53,463 records with successful feature extraction (Table 1).

Table 1: Dataset summary for the 53,463-record waveform audit. Duration is median [p05-p95].

Dataset	Records	Events	Stations	Catalog P	Duration (s)	Median M
InstanceGM	31,344	9,927	568	31,344	120.00 [120.00-120.00]	3.4
K-NET	22,119	1,528	921	21,141	119.00 [60.00-120.00]	4.3

86 InstanceGM contributes 31,344 records from 9,927 events and 568 stations. K-NET

contributes 22,119 records from 1,528 events and 921 stations. The two datasets have a median sampling rate of 100 Hz. InstanceGM records in the audit have a median duration of 120.00 s. K-NET records have a median duration of 119.00 s, with a 5th to 95th percentile range from 60 s to 120 s.

Records are grouped into low-magnitude background, M3-M4 small-event, and M4+ strong-motion strata (Table 2). InstanceGM contributes 9,000 low-magnitude background records, 9,000 M3-M4 records, and 13,344 M4+ records. K-NET contributes 21 low-magnitude background records, 5,647 M3-M4 records, 15,473 M4+ records, and 978 records with incomplete catalog timing for catalog-P comparison. The K-NET low-magnitude group is too small for a separate low-magnitude conclusion. It is retained in the full audit and reported as a data-availability note.

Table 2: Priority-stratum summary used in the waveform audit.

Dataset	Stratum	Records	Median duration (s)	Median M
InstanceGM	Low magnitude	9,000	120.00	2.2
InstanceGM	M3-M4	9,000	120.00	3.2
InstanceGM	M4+	13,344	120.00	4.4
K-NET	Low magnitude	21	60.00	2.9
K-NET	M3-M4	5,647	119.00	3.8
K-NET	M4+	15,473	119.00	4.5
K-NET	Other timing	978	119.00	4.4

K-NET waveforms are read from the full converted archive and high-pass filtered at 1 Hz before feature extraction. This preprocessing matches the earlier K-NET waveform representation used in the project and separates raw acceleration offsets from product-window behavior. Strong-motion filtering and baseline choices can alter downstream amplitudes and spectra, so preprocessing is kept explicit (Boore, 2001; Douglas and Boore, 2011). Catalog P times support comparator windows and diagnostics. The main selector uses waveform-derived candidates.

Methods

Waveform Features and Candidate Windows

For each three-component record, we form a vector-amplitude signal from the standardized component traces. The waveform-derived feature set contains an effective onset, an energy onset, an energy end, and significant duration. These features define five candidate window families:

1. Feature-onset fixed: from 2 s before effective onset to 40 s after effective onset, yielding a 42.00 s processing interval.
2. Energy-onset fixed: from 2 s before energy onset to 40 s after energy onset.
3. Catalog-P fixed: from 2 s before catalog P to 40 s after catalog P. This is an arrival-time comparator.
4. Adaptive energy-end: from 2 s before effective onset to 3 s after waveform energy end.
5. Full record: the complete record, used as the upper-bound processing interval.

The main selector considers feature-onset fixed, energy-onset fixed, and adaptive energy-end candidates. Catalog-P fixed windows remain in the audit as arrival-time-based comparators.

Product-Stability Criteria

Each candidate is evaluated against the full record. Let PGA_{full} and E_{full} denote the full-record peak vector amplitude and vector-signal energy. Let PGA_{win} and E_{win} denote the same quantities inside a candidate window. Let $t_{\text{peak,full}}$ denote the time of the full-record peak. A candidate is stable when all checks pass:

$$\frac{PGA_{\text{win}}}{PGA_{\text{full}}} \geq 0.99, \quad \frac{E_{\text{win}}}{E_{\text{full}}} \geq 0.95, \quad t_{\text{peak,full}} \in w.$$

The energy term follows the Arias-style idea of integrating squared motion over time, applied here as a relative vector-signal energy criterion (Arias, 1970). The threshold sensitivity analysis repeats the selector with alternative energy-retention criteria.

Shortest-Stable Selector

For each record, the selector filters candidate windows by the three product-stability checks, ranks the stable non-full candidates by duration, and chooses the shortest one. If the filter yields no stable non-full candidate, the selector assigns the full record. In compact form,

$$w^* = \arg \min_{w \in C_{\text{record}}} \text{duration}(w) \quad \text{subject to} \quad \text{stable}(w) = 1.$$

The selected window passes the stated audit by construction. The performance information comes from fixed-window failure rates, selected duration, candidate usage, full-record assignment, and product-impact changes relative to fixed-window baselines.

Response-Spectrum Retention Audit

We add an independent response-spectrum retention audit to connect processing windows with engineering strong-motion products. For each selected or fixed window, we compute 5% damping pseudo-spectral acceleration (PSA) at 0.2 s, 1.0 s, and 3.0 s and compare it with the full-record PSA (Chopra, 2017). A window has a PSA-retention failure when

$$\frac{PSA_{\text{win}}(T)}{PSA_{\text{full}}(T)} < 0.95.$$

This audit is separate from the selector. The selector uses PGA retention, relative energy retention, and peak-time inclusion; the response-spectrum check evaluates whether those selected windows also preserve period-dependent motion measures.

Results

Fixed Windows Are Dataset-Dependent

Fixed 42.00 s windows have different product-retention behavior in the two datasets (Figure 2; Table 3). InstanceGM fixed-window instability is high: 81.25% for feature-onset fixed windows, 69.91% for energy-onset fixed windows, and 73.03% for catalog-P fixed windows. K-NET fixed-window instability is much lower: 4.15%, 3.10%, and 8.25% for the same three windows. Under this audit, the same fixed duration acts as a dataset-dependent product-window policy.

The adaptive energy-end candidate reduces instability to 1.45% for InstanceGM and 0.23% for K-NET. It also exposes the scale difference between datasets: the median adaptive duration is 84.12 s for InstanceGM and 24.66 s for K-NET.

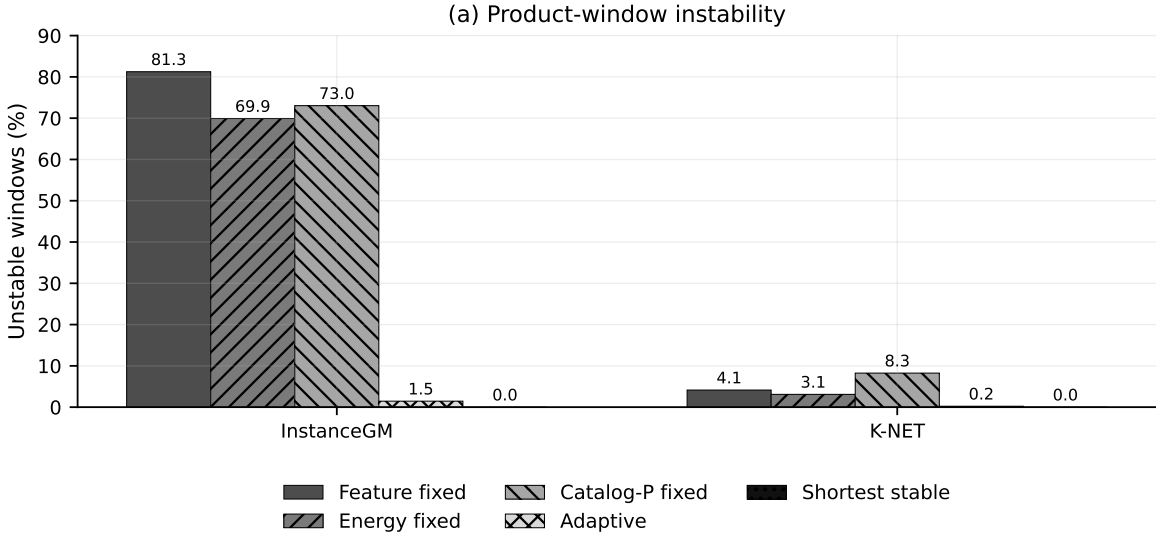


Figure 2: Product-window instability for fixed, adaptive, and selected windows on InstanceGM and K-NET records. Fixed 42.00 s windows fail frequently on InstanceGM and less often on K-NET, indicating dataset-dependent product retention. The shortest-stable selector removes product-retention failures under the stated audit because it filters candidates by the same criteria.

Magnitude Strata Preserve the Dataset Contrast

The M3-M4 strata show the same dataset contrast (Table 3). In InstanceGM M3-M4 records, feature-onset and energy-onset fixed windows have instability rates of 77.19% and 61.01%.

Table 3: Product-window stability summary for fixed, adaptive, and selected windows in the ALL priority group.

Dataset	Policy	Unstable (%)	Median window (s)	p05 energy	Full record (%)
InstanceGM	Feature fixed	81.25	42.00	0.080	0.00
InstanceGM	Energy fixed	69.91	42.00	0.228	0.00
InstanceGM	Catalog-P fixed	73.03	42.00	0.190	0.00
InstanceGM	Adaptive	1.45	84.12	0.953	0.00
InstanceGM	Shortest stable	0.00	84.94	0.954	1.31
K-NET	Feature fixed	4.15	42.00	0.957	0.00
K-NET	Energy fixed	3.10	42.00	0.963	0.00
K-NET	Catalog-P fixed	8.25	42.00	0.958	0.00
K-NET	Adaptive	0.23	24.66	0.960	0.00
K-NET	Shortest stable	0.00	24.66	0.960	0.18

The selector assigns 1.79% of records to the full interval and has a median selected duration of 71.14 s in the same stratum. In K-NET M3-M4 records, feature-onset and energy-onset fixed windows have instability rates of 1.63% and 1.61%. The selector assigns 0.30% of records to the full interval and has a median selected duration of 19.39 s. The windowing issue follows archive and waveform-scale behavior. Magnitude strata provide an internal check on that pattern.

Shortest-Stable Selection Preserves Products with Limited Full-Record Assignment

The shortest-stable selector assigns 0.84% of records to full-record processing under the default product-stability criteria (Figure 3; Table 3). Dataset-level full-record assignment is 1.31% for InstanceGM and 0.18% for K-NET. The full audit contains 451 records assigned to the full interval, including 412 from InstanceGM and 39 from K-NET. The median selected duration is 84.94 s for InstanceGM and 24.66 s for K-NET. The selected windows pass the three product-retention checks by design; assignment rate and duration summarize the operating cost.

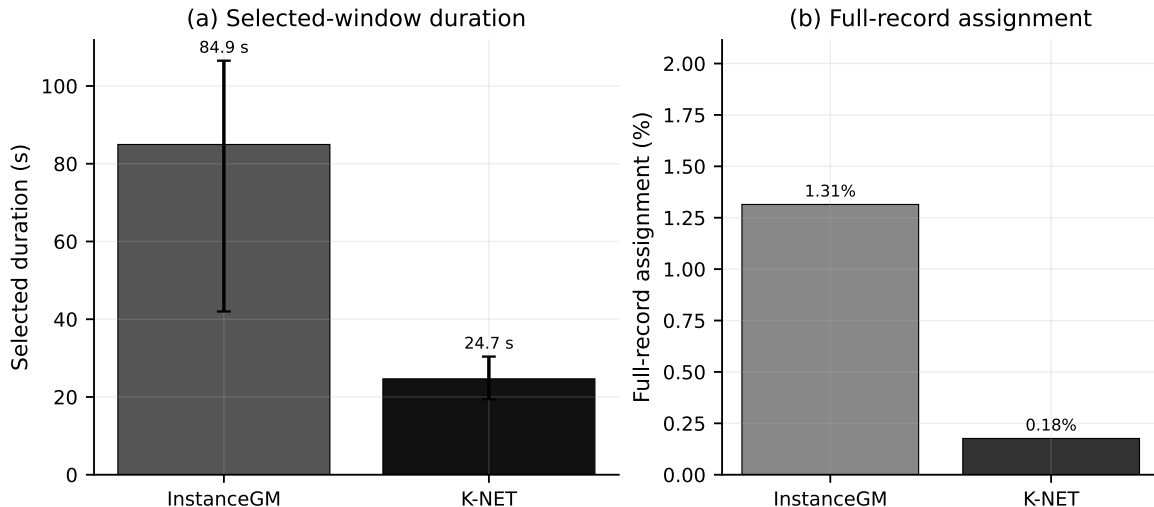


Figure 3: Selected-window duration and full-record assignment rate for the shortest-stable selector. The selector chooses longer typical windows for InstanceGM and shorter typical windows for K-NET. Only a small fraction of records are assigned to the full interval under the default PGA-retention and energy-retention criteria.

Candidate usage follows the dataset-scale difference. On InstanceGM, the selector chooses the adaptive energy-end candidate for 84.86% of records and the energy-onset fixed candidate for 13.52%. On K-NET, it chooses the adaptive energy-end candidate for 97.26% and the energy-onset fixed candidate for 2.44%. Full-record assignment occurs for 0.84% of all records.

Product Impact Relative to Fixed Baselines

The selector changes product retention and selected-window length relative to fixed baselines (Figure 4). On InstanceGM, it resolves 25,468 feature-onset fixed-window failures, 21,913 energy-onset fixed-window failures, and 22,889 catalog-P fixed-window failures under the product-retention audit. The median energy-retention gain relative to feature-onset fixed windows is 0.347, and the median duration change is 42.94 s. These values indicate that many InstanceGM fixed windows end before sufficient product-relevant energy is retained.

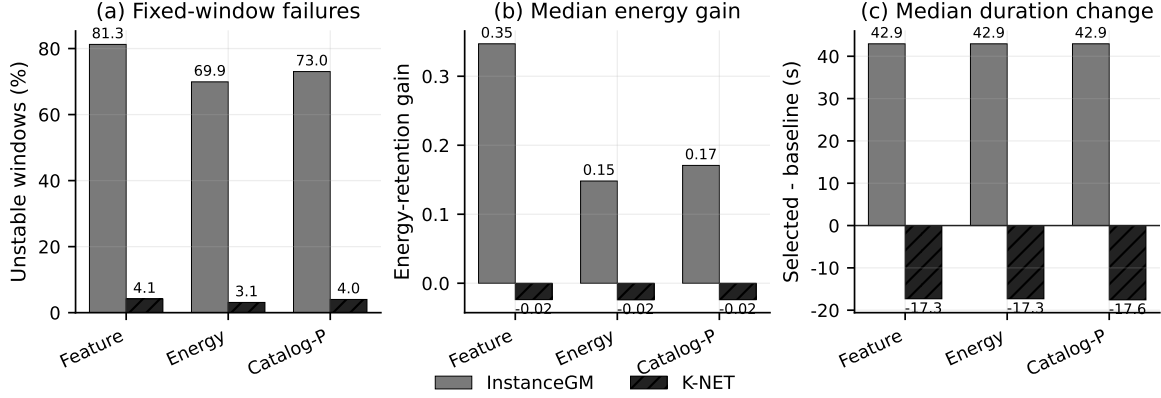


Figure 4: Product impact relative to fixed-window baselines. The panels show fixed-window instability, median energy-retention gain, and selected-minus-baseline duration change for feature-onset, energy-onset, and catalog-P fixed windows.

K-NET has fewer fixed-window failures. The selector resolves 917 feature-onset fixed-window failures, 686 energy-onset fixed-window failures, and 847 catalog-P fixed-window failures. The median duration change relative to feature-onset fixed windows is -17.31 s. The adaptive energy-end candidate often passes the product-retention checks with a shorter window than the 42.00 s fixed baseline.

Threshold Sensitivity

The selector is sensitive to the energy-retention criterion (Figure 5). With $PGA \geq 0.99$ and $E \geq 0.95$, the overall full-record assignment rate is 0.84%. Tightening energy retention to 0.98 raises the rate to 46.84% overall. InstanceGM full-record assignment increases from 1.31% at 0.95 energy retention to 71.84% at 0.98. K-NET assignment increases from 0.18% to 11.42%. The sensitivity analysis makes this trade-off explicit: stricter energy retention yields a more conservative selector and substantially more full-record assignments.

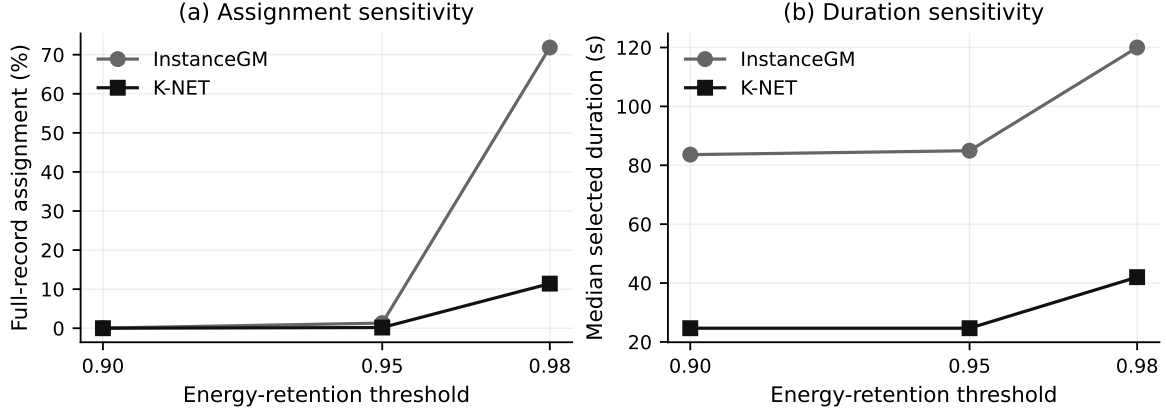


Figure 5: Sensitivity of the shortest-stable selector to the energy-retention criterion with the PGA-retention criterion fixed at 0.99. The default 0.95 energy-retention criterion keeps full-record assignment rare. A stricter 0.98 energy-retention criterion substantially increases full-record assignment, especially for InstanceGM.

Response-Spectrum Retention

The response-spectrum audit strengthens the product-window interpretation (Figure 6; Table 4). Across all 53,463 records, feature-onset fixed windows have PSA-retention failure rates of 12.98%, 22.26%, and 32.28% at 0.2 s, 1.0 s, and 3.0 s. The shortest-stable selector reduces the same rates to 0.02%, 0.87%, and 5.56%, corresponding to 9, 465, and 2,972 records.

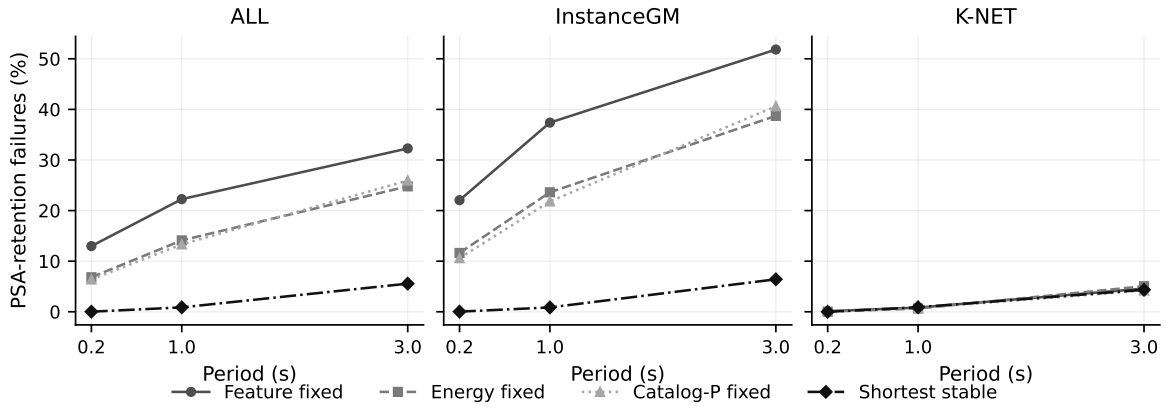


Figure 6: Response-spectrum retention at 5% damping. Panels compare PSA-retention failure rates at 0.2 s, 1.0 s, and 3.0 s for fixed windows and the shortest-stable selector. The selected windows reduce overall PSA-retention failures from 12.98%, 22.26%, and 32.28% for feature-onset fixed windows to 0.02%, 0.87%, and 5.56%.

The 3.0 s period exposes the main residual risk. On InstanceGM, feature-onset fixed windows have 51.83% PSA-retention failures at 3.0 s, and the selector has 6.42%. On K-

Table 4: Response-spectrum retention summary. Values are 5% damping PSA-retention failure percentages relative to the full record.

Dataset	Policy	0.2 s failure (%)	1.0 s failure (%)	3.0 s failure (%)
ALL	Feature fixed	12.98	22.26	32.28
ALL	Shortest stable	0.02	0.87	5.56
InstanceGM	Feature fixed	22.06	37.39	51.83
InstanceGM	Shortest stable	0.03	0.86	6.42
K-NET	Feature fixed	0.11	0.83	4.59
K-NET	Shortest stable	0.00	0.89	4.34

NET, the same comparison is 4.59% and 4.34%. The selector substantially improves long-period retention for InstanceGM without increasing K-NET losses. This result clarifies the engineering relevance of the spectrum audit: short-period and intermediate-period products are largely preserved under the selected windows, while long-period PSA remains the product most likely to expose residual windowing loss. Records failing the 3.0 s PSA threshold are listed in the record-level audit material for long-period structural-response or soft-site product review.

Discussion

The results show that strong-motion processing windows need product-retention checks together with selected-window length. InstanceGM and K-NET show distinct window scales under a common audit. A fixed 42.00 s window is too short for many InstanceGM records and longer than needed for many K-NET records. The product-check selector adapts to that difference, and its operating cost is summarized by full-record assignment and selected duration.

The method leaves a clear audit trail. Each selected window can be traced to its candidate type, product-retention values, peak-inclusion status, and full-record assignment state. This record-level traceability matters for strong-motion product preparation because processing decisions need to survive later review and archive updates.

The record-level audit packet connects the aggregate results to practical review. It in-

cludes examples where fixed windows miss late product-relevant motion and the selector restores retention, examples where K-NET records keep the product checks with compact windows, and boundary cases assigned to the full record. These cases let an operator inspect why a record received its processing window without using hidden labels or manual arrival picks.

These results support an offline product-window policy. The full record is available before the processing window is finalized, and each selected window is evaluated against full-record products. The 0.95 energy-retention criterion is a design parameter supported by the sensitivity analysis. A stricter 0.98 criterion pushes many InstanceGM records to full-record processing. The response-spectrum audit adds an engineering-product check and shows that the selector retains most 0.2-3.0 s PSA values, with the largest residual loss at 3.0 s. The remaining 3.0 s failures identify records that need conservative handling for long-period spectral products. The selected windows still give a large product-level gain at 0.2 s, 1.0 s, and most 3.0 s records. Future work can start from this explicit product-stability baseline and test group-conditioned thresholds on an external validation archive.

Conclusions

We evaluated product-stable window selection for 53,463 strong-motion records from InstanceGM and K-NET. Fixed 42.00 s windows show strong dataset dependence, with high instability on InstanceGM and much lower instability on K-NET. The shortest-stable selector uses waveform-derived candidates and product-retention checks to choose compact processing windows. Under the default audit, full-record assignment is 0.84% overall, while median selected duration differs sharply by dataset: 84.94 s for InstanceGM and 24.66 s for K-NET. The response-spectrum audit shows that overall PSA-retention failures drop from 32.28% for feature-onset fixed windows to 5.56% for the selected windows at 3.0 s. The resulting contribution is an auditable offline product-window policy with explicit thresholds, full-record

assignment accounting, cross-dataset sensitivity, and engineering-product retention checks.

Data and Resources

Waveforms from the InstanceGM/INSTANCE data family and K-NET were used in this study. K-NET waveforms were converted with explicit UD $\rightarrow$ Z, NS $\rightarrow$ N, and EW $\rightarrow$ E component mapping, and K-NET waveform features were computed after 1 Hz high-pass preprocessing. The reproducibility release contains the source code, focused tests, manifest and worklist files, waveform-feature summaries, window-stability summaries, selector summaries, product-impact summaries, threshold-sensitivity summaries, response-spectrum audits, record-level audit cases, figure sources, checksums, and command log. The public release is archived at <https://github.com/zhouhaoyiu/strong-motion-product-window-qc/releases/tag/v0.1.0>. InstanceGM/INSTANCE data were accessed through <https://doi.org/10.13127/INSTANCE> on 16 June 2026. K-NET/NIED data were accessed through <https://doi.org/10.17598/NIED.0004> on 16 June 2026. Raw waveform archives are not redistributed and remain subject to provider terms. Code and focused tests are released under the MIT License; derived summaries, figures, record-audit plots, manuscript-support metadata, and documentation are released under CC BY 4.0.

An AI-assisted language editing tool was used only to polish manuscript wording. The authors checked the scientific content, numerical outputs, source code, figure data, and final manuscript.

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Declaration of Competing Interests

The authors declare no competing interests.

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